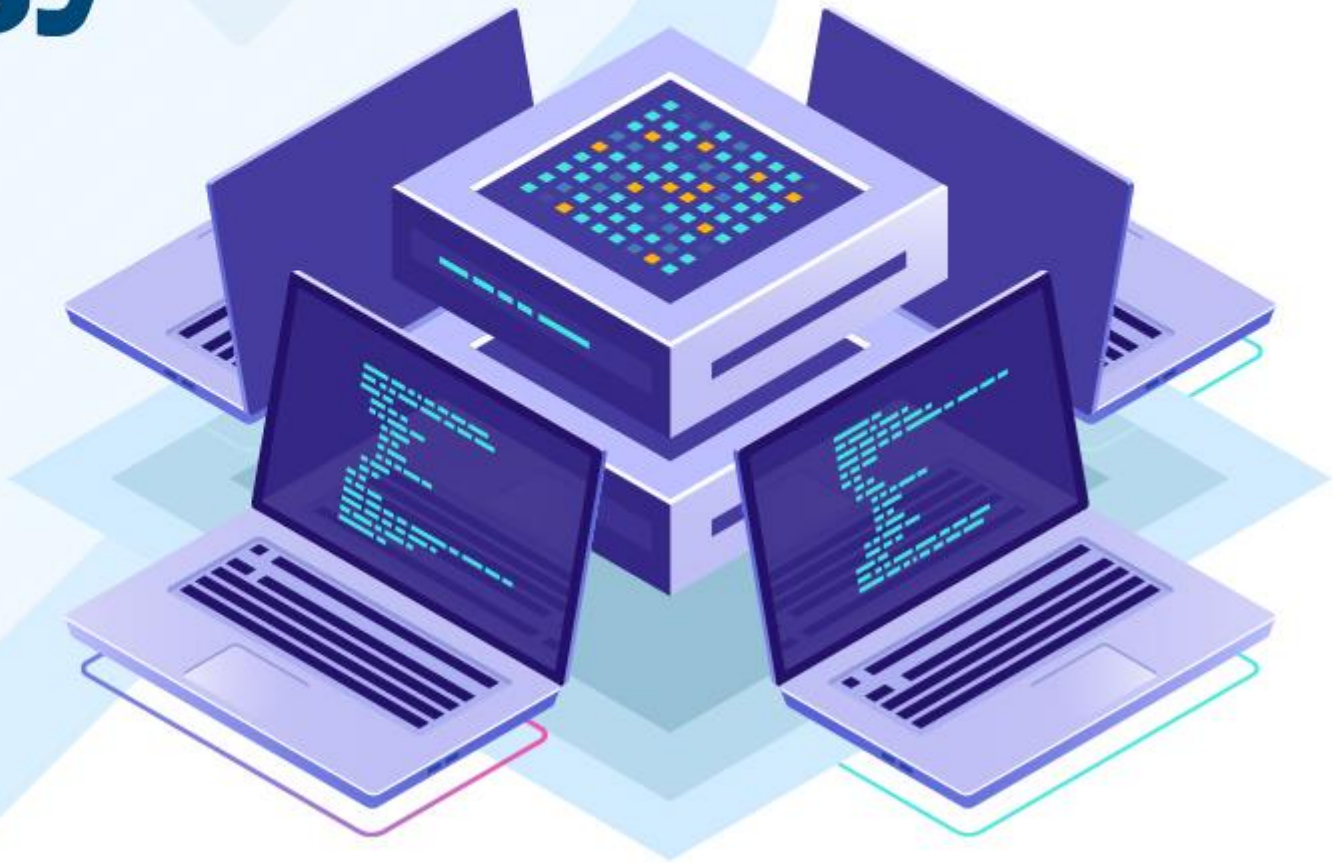
Information Technology

Fall Semester
2026 - 2027

- Embedded Systems
- **Dr. Yasmine Mahmoud**



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Register Files

- Specific Purpose Registers (SPR):

  5. Process Status Word (PSW):

  - Tt is a set of flags; each flag is a bit and has a specific function.

  Ex:

  - (Sign flag) Bit represent the sign of the last result from ALU.
  - (0 means :+ve , 1 means -ve)
  - SO, last instruction executed in ALU, its result is now stored in these flags.

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Register Files

- Specific Purpose Registers (SPR):
  - SO, last instruction executed in ALU, its result is now stored in these flags.
  - But lots of instructions is not executed in ALU. Ex. Jump instruction (JMP), Increment instruction (x++) (INC)
  - X++ will be converted into INC instruction in assembly **while x is in memory, not fetched in ALU**.

Register Files

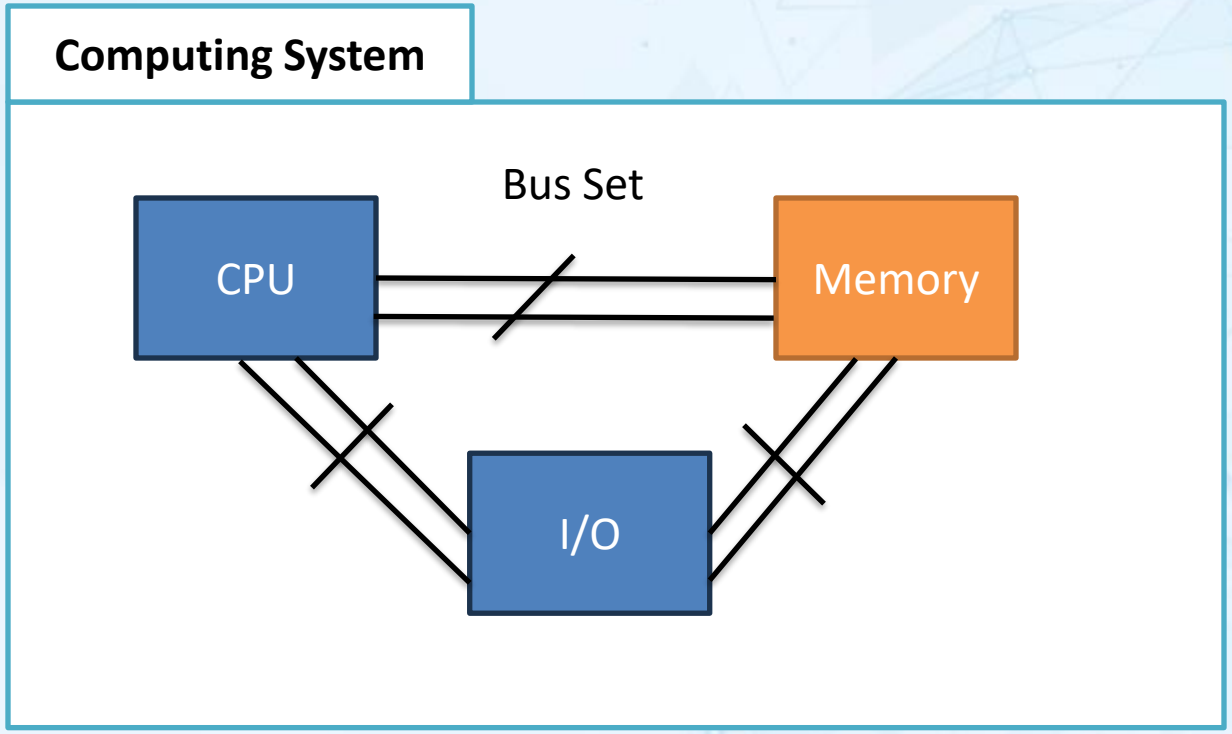
- Specific Purpose Registers (SPR):
  - Jump instructions like **JMP:** alter the flow of execution but do not perform any arithmetic or logical computation. Their role is to change the **Program Counter (PC)** to point to a new address in memory.
  - No ALU operation is required because no computation is performed.

Register Files

- Specific Purpose Registers (SPR):
  - The increment instruction (often written as INC in assembly) is used to add 1 to a register or memory location. Although this operation involves an arithmetic operation, it does not always require the ALU, depending on the specific architecture.
  - Even though INC is an arithmetic operation (adding 1), **it might not involve the ALU in certain processor designs**. Some processors have specialized hardware that can perform simple operations like incrementing a register without invoking the ALU.

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Memory



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Memory

- It is a set of locations, each location size vary based on the architecture.
- Ex. Microprocessor 8-bits architecture, which means that width of the memory register is 8-bits.
- Also means that the two operands in ALU, max size is 8-bit.



Memory

- If the two operands is larger than 8-bits. Ex 16-bit??
- So, Slicing is the solution. Which means to divide the bits to more than one operation.

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Memory

- 8-bit architecture, so each location is 8-bit.
- Location is 1 byte (8-bits).
- Each bit holds 0 or 1 value.

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Memory

- Access the memory means read or write on it.
- Access time: it is an important factor in memory.
- Write (w) access time is bigger than read (r) access time.
- Note that, the more time passes, the slower memory speed gain.

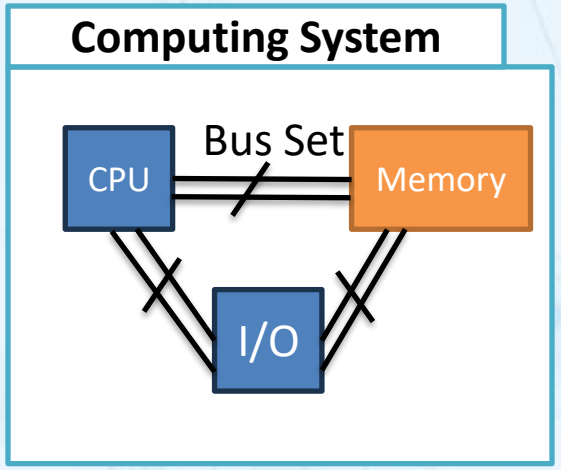
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Memory

- Each bit is consists of D-flip-flops.
- Each 8 D-flip flops construct 8-bit register.

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Memory



- How to read or write in memory?
- CPU & Memory are connected using Bus set
- (data bus, address bus, & control bus)

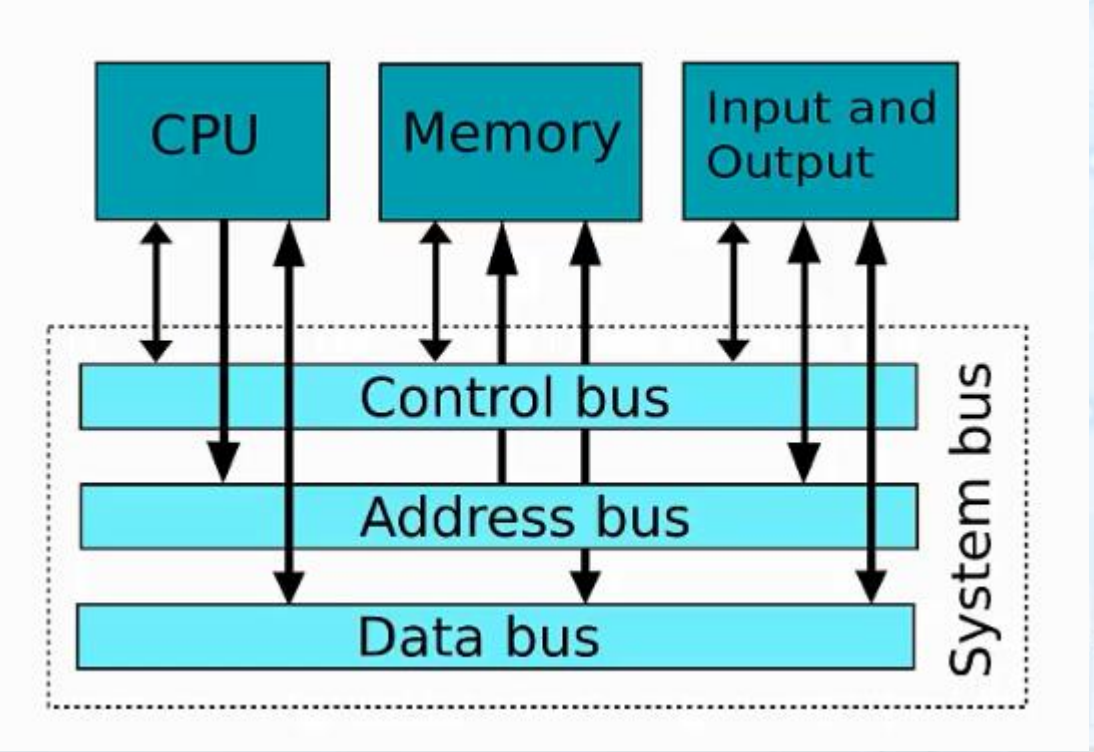
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Memory

- There are normally three types of bus in any processor system:
  - **An address bus:** this determines the location in memory that the processor will read data from or write data to.
  - **A data bus:** this contains the contents that have been read from the memory location or are to be written into the memory location.
  - **A control bus:** this manages the information flow between components indicating whether the operation is a read or a write and ensuring that the operation happens at the right time.

Memory

- Bus size:
- Bus is not one wire; it is a set of wires maybe 16 wire or 32 as an example.

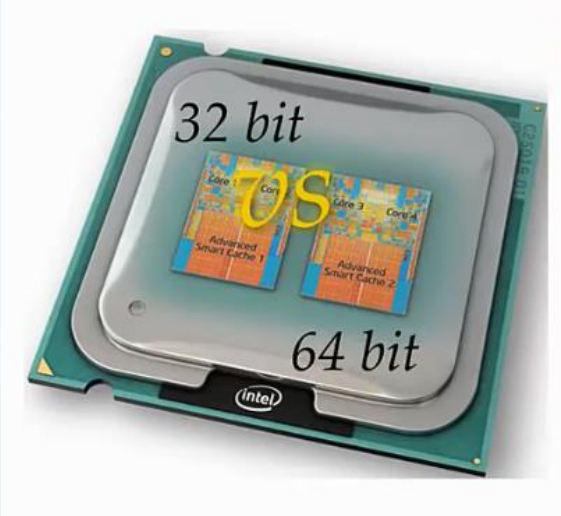


Memory

- Bus size:

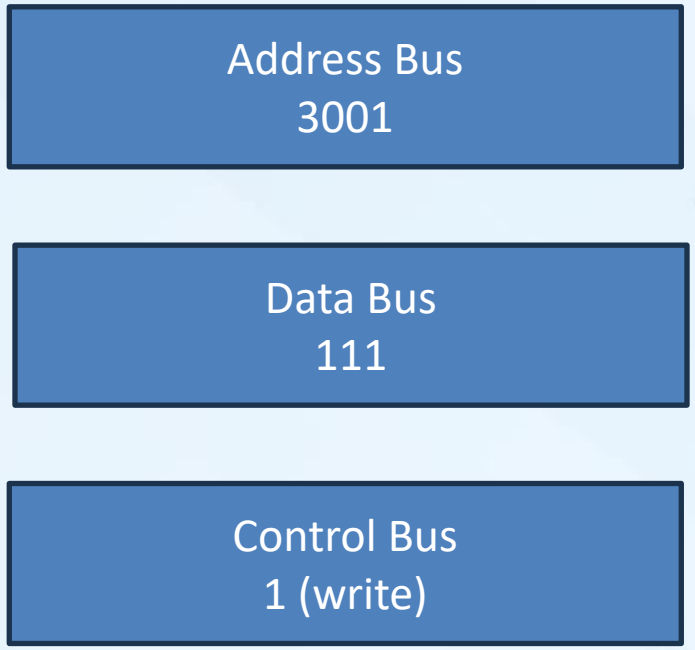
It differ in no. data to be transferred at once.

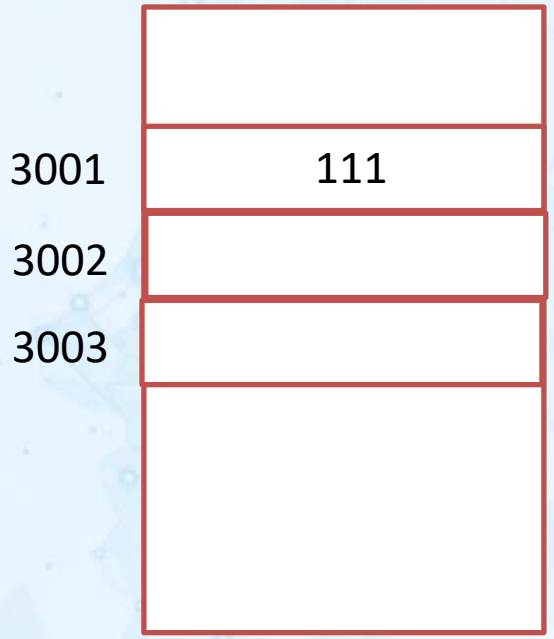
So, we know that there is 32-bit pc or 64-bit pc.



Load & Store Operations (Read/Write)

- Example





Read and Write operation time

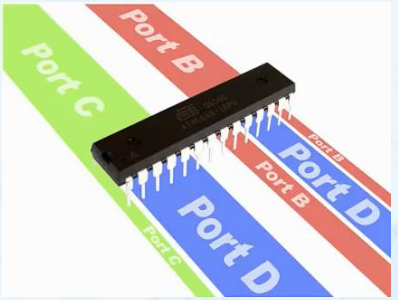
- Read time is less than write time, because read operation is just read, but write operation may have to clear data in memory first then write.

Three key advantages of memory:

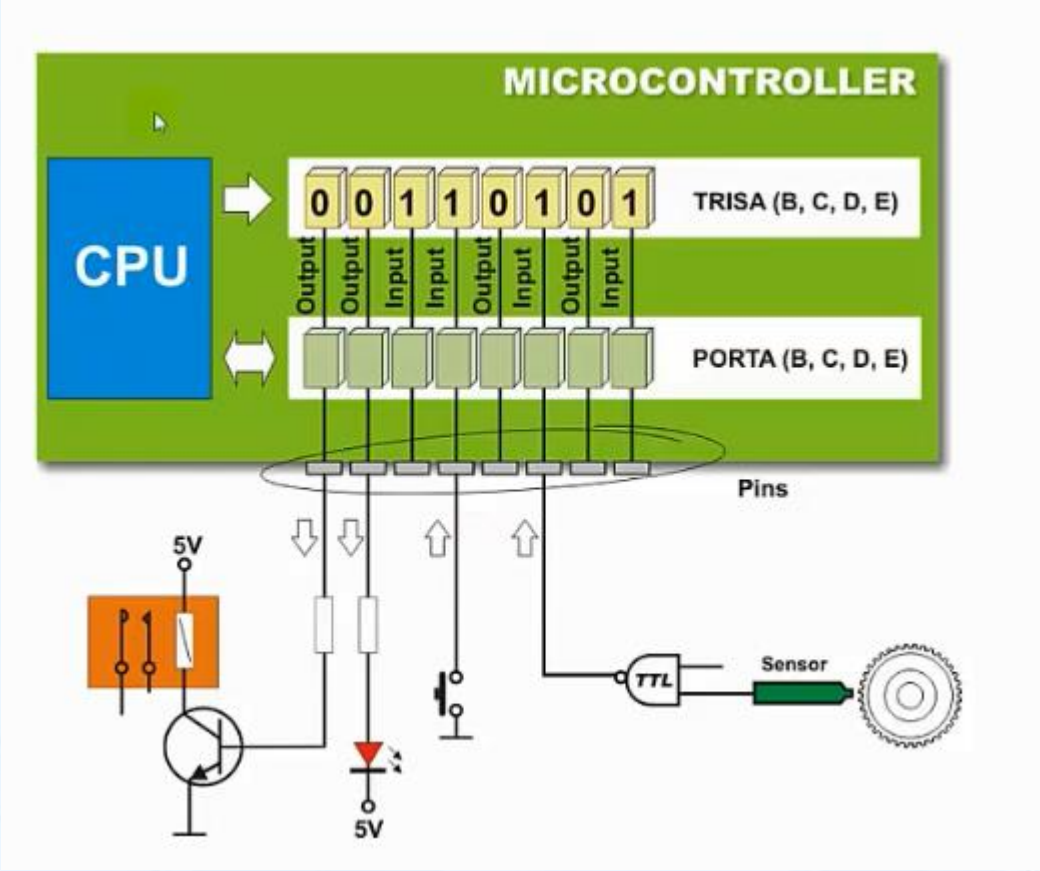
1. Capacity : How many bit? (Memory Size)
   - Address line in address bus determine memory size.
2. Speed / Access time
   - Read or Write
3. Organization
   - If you want 16 K memory, but you have only 4 K, memory is organized then you can connect four 4K , 4K , 4K , 4K it gives 16K.

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Port



- Port in microcontroller, allow us to deal with the external word.
- Ex. To deal with led, transistor, …
- CPU need port to deal with external word.

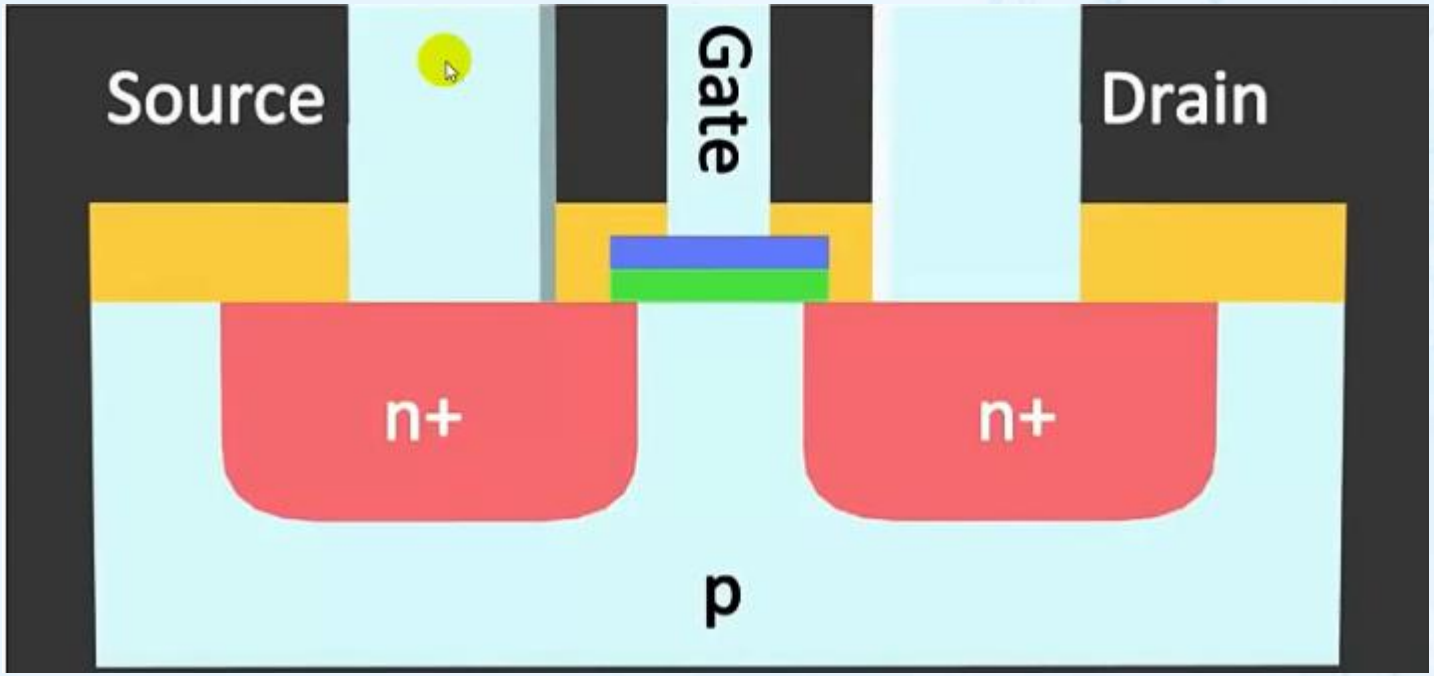


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Logic Operation

- Transistor (MOSFET)





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Logic Operation



- Transistor (MOSFET)
- Consists of three pins:

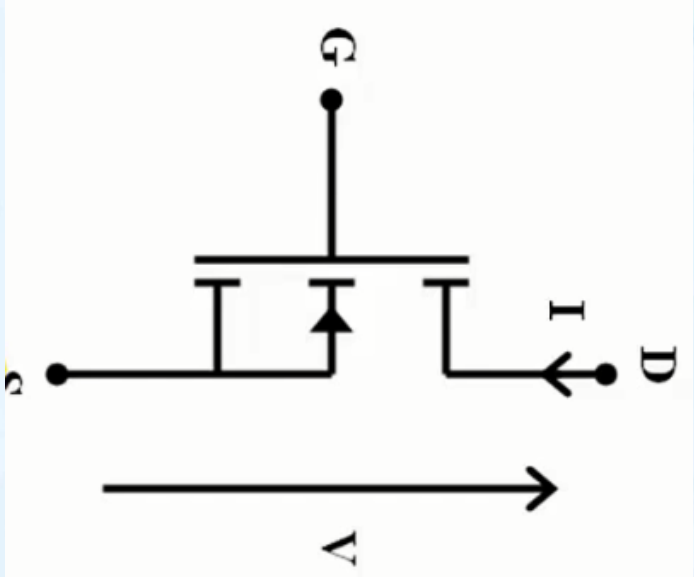
  (Source, Gate, and Drain)
- It is n-type as it contains n-regions, each region contains electrons.
- Our aim is to transfer from Source electric current into Drain.

Logic Operation

- Transistor (MOSFET)
- Consists of three pins:
  (Source, Gate, and Drain)

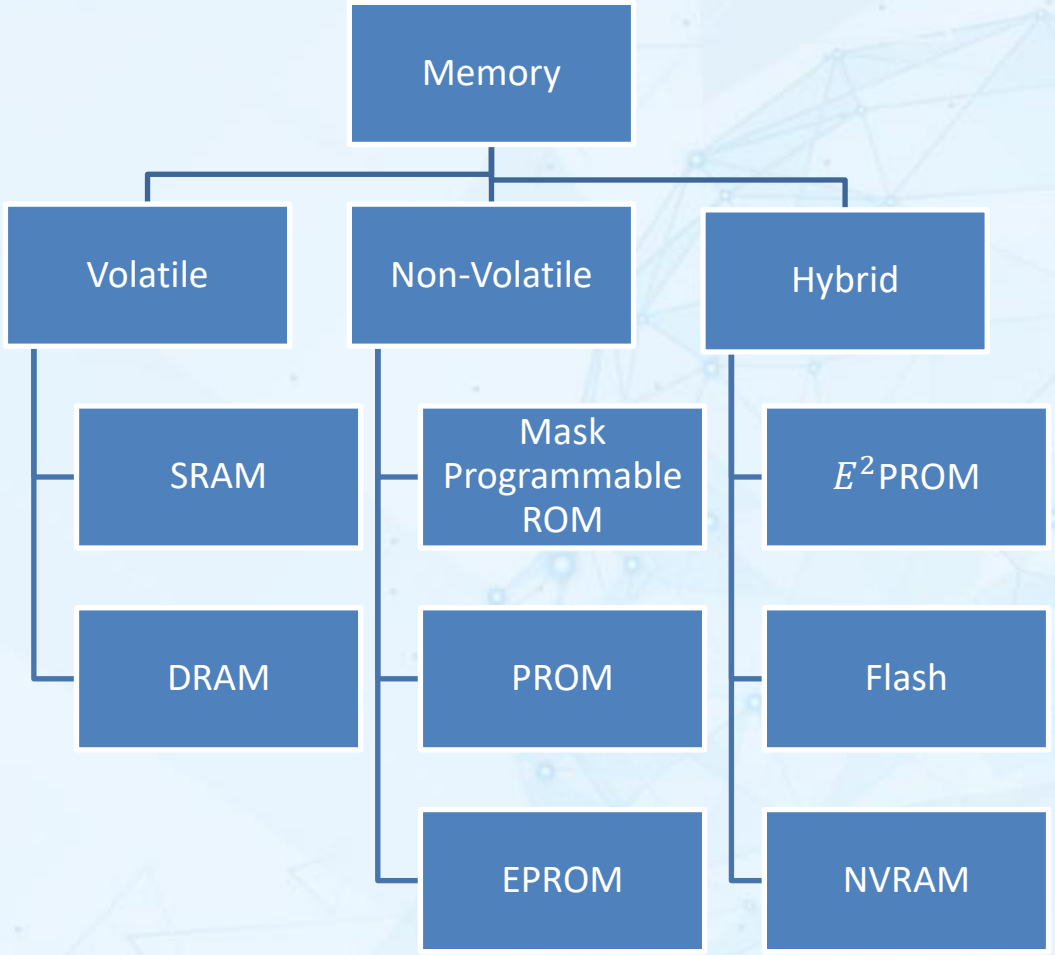






Memory Types

1. Three types of memory:



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THANK YOU FOR WATCHING

QUESTIONS?

